

Deep Full-scaled Metric Learning for Pedestrians Re-identification

A Pre-requisite Study on Multi-camera-based Affective Computing

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ABSTRACT

In this study, a new full-scaled deep discriminant model is proposed to tackle the re-identification (re-id) problem of pedestrian targets, which aims to identify pedestrian targets within a network of cameras with non-overlapping fields of view and is pre-requisite in multi-camera-based affective computing. The new full-scaled model is realized by taking concepts of depth, width, and cardinality simultaneously into consideration, and the challenging re-id problem in this study is further tackled via a novel deep semi-supervised metric learning method based on the full-scaled model. Additionally, both the conventional stochastic gradient descent algorithm and an alternative more efficient proximal gradient descent algorithm are derived to realize the new deep metric learning method. For experimental evaluations, the novel full-scaled deep metric learning method has been compared with 9 other popular re-id methods based on 3 well-known databases. Comprehensive statistical analyses suggest the superiority of the new method when handling the balance learning problem in the re-id task.

CCS CONCEPTS

• Computing methodologies; • Artificial intelligence; • Computer vision;

KEYWORDS

Deep Full-scaled Metric Learning, Pedestrians Re-identification, Multi-camera Affective Computing

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1 INTRODUCTION

Affective computing is an emerging field which receives more and more research attention in recent years. When multiple cameras are equipped to locate the pedestrian target and in turn analyze her/his affective expressions based on meta-data in terms of video and audio acquired from multi-view cameras, the above important affective computing problem actually becomes a challenging multi-camera-based affective computing problem [20]. Also, it is intuitive to perceive that, accurately identifying the pedestrian target via a specific individual camera and then re-identifying (i.e., re-id) the same target again across a network of different cameras with non-overlapping fields of view becomes essential and pre-requisite, since it is essential for post-hoc affective computing to be executed on correctly identified pedestrians. The general process of the above challenging re-id problem can be summarized as follows. When a pedestrian target is determined within a camera view, an image patch eliminating the main background and highlighting the pedestrian target will be selected. After that, extracted features representing attributes of the pedestrian target will be constructed. Then, the same type of features extracted from image patch candidates of other camera views will be evaluated, so that the one with the highest feature similarity will be shortlisted as the re-id outcome [20]. Based on the above description, it is clear that, both proper features and suitable similarity measures play important roles in solving the challenging re-id problem.

Generally speaking, the main progression of solving the re-id problem complies well with the advance of machine learning. In the era of shallow learning, conventional hand-crafted low-level features motivated by senior researchers' expertise are quite valuable. For instance, in [21], histograms of color and texture are employed as subject-discriminative features, and their capabilities in differentiating various identities are revealed via comprehensive experiments in that study. In [3], the specific re-id problem in airport scenarios is concentrated, and its hand-crafted features are made up of the popular KLT (i.e., Kanade-Lucas-Tomasi) optical flow, color and texture histograms. The effectiveness of combining above hand-crafted features is suggested from that study as well. In [19], depth images, which are popular in many computer vision studies, are applied so that depth voxel covariance descriptors can be constructed as hand-crafted low-level features in that study. In [26], the popular low-rank and sparse representation in the era of shallow learning is employed to make the hand-crafted low-level feature more effective and efficient for solving the re-id problem.

Based on above introduced studies utilizing hand-crafted features to solve the re-id problem, it can be summarized that, low-level features, such as color, texture, optical flow, depth, etc., are heavily relied on and the indispensable expertise of senior researchers are highly required.

When deep learning becomes the current main trend of machine learning beginning from the last few years, the heavy burdensome of delicately designing hand-crafted low-level features can be substantially alleviated via automatically learned latent features from diverse deep learning models through their end-to-end back-propagation learning procedures. Also, it is widely acknowledged that, deep learning receives vast popularity accompanying with the time when powerful computational hardware (e.g., GPU, FPGA, customized processing units, AI chips, etc.) become more and more affordable for ordinary ending users in recent years. Generally speaking, most contemporary deep learning models can be categorized into two kinds, i.e., deep generative learning models and deep discriminant learning models. Deep generative learning models can be viewed as deep layer-based models, whose main purpose is to replicate the data or its distribution that ending users would like to have (i.e., the named “fake-but-realistic” data from real data). Two major deep generative learning models that have been vastly utilized are AE (i.e., auto-encoder) and GAN (i.e., generative adversarial networks) [5]. For AE, it simply adopts hidden layers, in which less neurons are contained than the input layer, to generate compressed data that can be reconstructed as close as possible towards the input data. GAN, on the contrary, is more sophisticated and becomes frequently utilized at the current stage in deep learning. GAN is often made up of two component sub-networks, one is the generator sub-network, the other is the discriminator sub-network. The role of the generator sub-network is to generate fake-but-realistic data, and the role of the discriminator sub-network is to differentiate fake-but-realistic data from real data. When the training of GAN proceeds, the capability of generating fake-but-realistic data in the generator sub-network as well as the capability of differentiating fake-but-realistic data from real data in the discriminator sub-network will be boosted, simultaneously.

Deep discriminant learning model, on the other hand, focuses more on discrimination / classification utilizations since it utilizes softmax as the simple mapping way to realize its classification. Generally speaking, well known deep discriminant learning models are often winners of noticeable competitions in vision-based studies in recent years (e.g., the well-known ILSVRC [13], etc.), and they include but not limit to AlexNet [10], VGG [15], GoogleNet [17], ResNet [7], etc. Besides the increasing depth of contemporary deep discriminant learning models, their another interesting trend is that, model structures become more and more sophisticated. There are several key points to be addressed here. First, the width of deep discriminant learning models largely increases. In [25], it is reported that a wide 40-layer ResNet model can obtain the similar generalization capability as a conventional “narrow” single-channel 1001-layer ResNet model, but the wide model only costs 1/8 training time of the narrow one. Second, the cardinality of deep learning models largely increases. Cardinality is generally regarded as the number of isolated paths in a deep learning model, and all paths are required to share the same topology structure [22]. Many current deep discriminant learning models follow up with the above idea

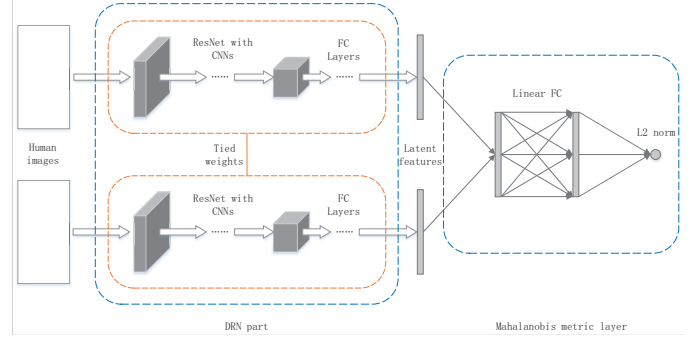


Figure 1: The main structure of the new full-scaled deep learning model for solving the re-id problem in this study.

and employ multiple channels to realize their cardinalities. Third, novel structures of deep learning models, such as the capsule [14], are also newly investigated and proposed.

In this study, a novel deep discriminant learning model simultaneously taking concepts of depth, width, and cardinality into consideration (i.e., a “full-scaled” structure) is introduced to tackle the re-id problem, which is pre-requisite and essential in multi-camera-based affective computing. Also, a new deep metric learning method, i.e., the deep full-scaled metric learning, is proposed based on the new full-scaled model. Generally speaking, metrics can be viewed as a special type of distances, in which four axioms (i.e., non-negativity, identity of indiscernible, symmetry, and triangle inequality) need to be met at the same time. Most contemporary deep metric learning methods heavily rely on well-established deep learning models [12]–[24]. However, this study is significantly different from them by introducing the novel deep metric learning method based on the new full-scaled deep discriminant learning model.

The organization of this paper is as follows. In Section 2, the structure of the new full-scaled deep learning model is elaborated. Details of its new deep metric learning method is introduced. The new method is realized by both a conventional stochastic gradient descent (SGD) algorithm in which the gradient has been derived and a more efficient proximal gradient descent (PGD) algorithm whose details are also elaborated. In Section 3, totally 10 deep learning-based / shallow learning-based methods are compared based on 3 popular re-id databases for comprehensive experiments. Comprehensive analyses are carried out from the statistical point of view to suggest the superiority of the new method. In Section 4, the conclusion of this paper is drawn.

2 FULL-SCALED DEEP METRIC LEARNING

In this section, the new full-scaled deep metric learning method for solving the re-id problem is elaborated. It is necessary to point out that, the new full-scaled deep learning model is inspired by the original single-channel ResNet model [8], but significantly different from the original model as concepts of depth, width, and cardinality are all highlighted in this new full-scaled model. To be specific, the depth concept suggests that the full-scaled model should have

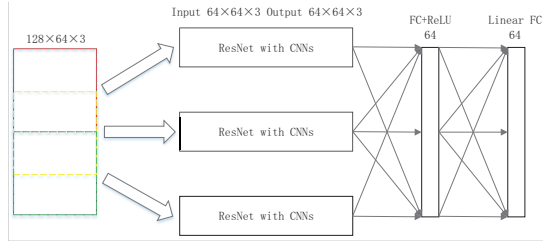


Figure 2: The three-branch structure in each individual channel of the new full-scaled model.

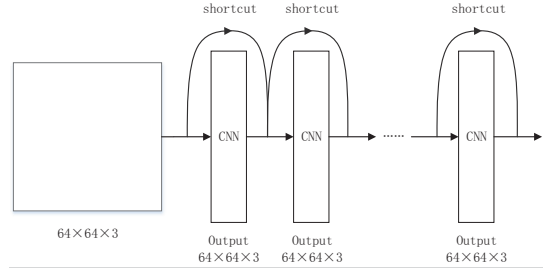


Figure 3: The structure of “ResNet with CNNs” in each individual branch of Figure 2.

a tremendously deep structure in order to achieve good generalization performance, which is similar towards other conventional deep learning models. The width concept implies that the full-scaled model should have a wide model structure. The purpose to increase the width of the full-scaled model is because that, training very deep ResNet models may result in a serious problem of diminishing feature reuse, and it has another serious drawback of slow trainings. In order to overcome the above difficulties, wide ResNet models are necessary. The cardinality concept requires the structure of the new full-scaled model to be homogeneous and multi-channel-based, so that the accuracy of the model can be further improved. After having all above issues taken into consideration, the main structure of the new full-scaled model in this study is demonstrated in Figure 1. It can be noticed from Figure 1 that, the new model has a twin-channel structure, which enables a pair of re-id images to be inputted simultaneously. The reasons are as follows. First, the concept of cardinality can be easily realized via the twin-channel structure. Second, the twin-channel structure allows the same parameters setting (i.e., tied weights) in each individual channel to be shared, which is more efficient for the model’s training. Third, the following deep metric learning procedure can be fulfilled as a semi-supervised deep metric learning in this study. It is also necessary to highlight in Figure 1 that, the full-scaled model is mainly revealed as the DRN (i.e., deep residual networks) part, and deep latent features automatically determined from it will be fed into a following Mahalanobis metric layer for realizing the deep metric learning.

Details of the full-scaled model are illustrated from Figures 2 to 4. Within each individual channel of the new model, there are three branches, whose structure is shown in Figure 2. Details of

constructing such a three-branch structure in each channel are as follows (i.e., specific image sizes in the following descriptions are illustrated based on the well-known CUHK01 database. Sizes may vary regarding other databases). First, provided the spatial resolution of an input image in this study as $128 \times 64 \times 3$, 3 overlapping image patches whose spatial resolutions are consistently set as $64 \times 64 \times 3$ can be obtained from the original input. Then, each individual patch of them will be fed into one of the 3 branches independently. Within each branch of Figure 2, the main structure follows the idea of the ResNet model, whose non-linear mapping is realized via CNN (i.e., the above procedure is annotated as “ResNet with CNNs” in Figure 2). In Figure 3, such a detailed structure of “ResNet with CNNs” in each individual branch of Figure 2 is further elaborated. There are also several issues to be addressed here. First, CNN acts as a non-linear stacked weighted mapping, whose details are demonstrated in Figure 4. Second, weights of CNNs among three branches cannot be shared, as three branches emphasize on different spatial areas of the input image. Third, in each branch of Figure 3, a convolution kernel of the size 1×1 is employed to realize the shortcut as an identity mapping, and outputs of three branches in Figure 2 are directly connected towards a FC layer with ReLU as the activation function. It is necessary to point out that, in the CNN component of the new full-scaled model (i.e., Figure 4), a series of operations are conducted to increase the width of the new model. To be specific, after the input image patch (i.e., with the spatial resolution $64 \times 64 \times 3$) is fed into the CNN component, a double operation is executed to duplicate the feature map, whose outcomes will both undergo through the first convolution (i.e., Conv.1 with a 11×11 kernel and stride 1) operation and then concatenate (i.e., concat) before the second convolution operation is carried out (i.e., Conv.2 with a 1×1 kernel and stride 1). In this way, the number of feature maps can be increased, and the width of the new model can be enlarged as well. Besides the above duplication / concatenation operations to increase the model width, the multi-channel structure in Figure 1 as well as the multi-branch structure in Figure 2 are also effective to increase the width of the new full-scaled model.

2.1 Full-scaled Deep Metric Learning via SGD

In this study, the Mahalanobis metric $d(x_1, x_2) = \sqrt{(x_1 - x_2)^T M (x_1 - x_2)}$ is emphasized, in which the covariance matrix M is to be learned. According to general requirements of being a metric, the covariance matrix M needs to be symmetric and PSD. Thus, M can be further decomposed as $M = WW^T$ via the classic Cholesky factorization. Hence, the original Mahalanobis metric can be further derived as Equation 1.

$$d(x_1, x_2) = \sqrt{(x_1 - x_2)^T WW^T (x_1 - x_2)} = \sqrt{(W^T (x_1 - x_2))^T (W^T (x_1 - x_2))} = \|W^T (x_1 - x_2)\|_2 \quad (1)$$

in which, $W^T (x_1 - x_2)$ can be realized via the linear FC layer illustrated in Figure 1. Therefore, the Mahalanobis metric learning in this study can be viewed as learning matrix W in the weighted Euclidean metric of Equation 1.

Since the full-scaled deep metric learning in this study is semi-supervised, the notion of side-information is necessary to be constructed and it is realized based on the Kendall-Tau correlation coefficient measurement (i.e., KT measure) in this part. The original

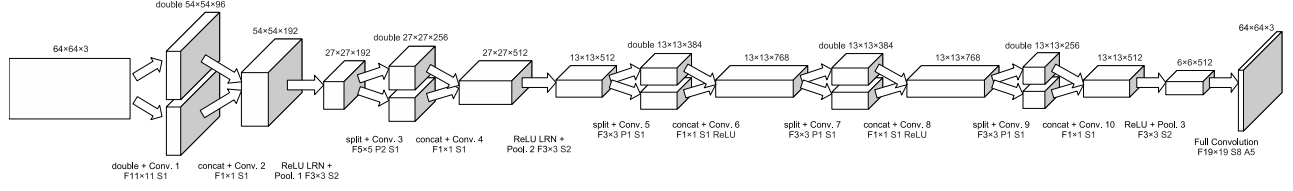


Figure 4: The detailed structure of the CNN component in Figure 3.

definition of the KT measure is explicitly described in Equation 2.

$$KT = \frac{N}{N_n} = \frac{P - Q}{N_n} = \frac{P - Q}{\frac{1}{2}n(n-1)} \quad (2)$$

where, P and Q denote concordant pairs and discordant pairs that are utilized to construct a pair of constraints to represent the side-information in this semi-supervised deep metric learning; N_n is a normalization term controlling the range of the KT measure within $[-1, 1]$. The meaning of terms P and Q can be interpreted via the ranking perspective. Provided an image pair (x, y) in which image x is considered to be more similar towards a given query image than image y in a ranking images list. If the ranking images list is sorted according to a decreasing similarity order and image x is ranked before image y in such a list, (x, y) will form a concordant pair. Otherwise, (x, y) forms a discordant pair. It is also valuable to mention that, P and Q are like similar and dissimilar pairs in the well known semi-supervised metric learning study of the shallow learning era [23], except that P and Q in this study are explicitly defined based on both notion of similarities and ranking positions.

In order to fulfill deep metric learning based on Equation 2, an approximation process is necessary. The main purpose is to change the discrete KT measure into a surrogated continuous one, as discrete characteristics coming from terms P and Q prevent the conventional optimization process to be executed on Equation 2. Therefore, terms P and Q are first re-written as Equations 3 and 4.

$$\text{Concordant pair (P):} \quad \text{sgn}(s_{(x,y)}) \cdot \text{sgn}(\ell_{(x,y)}) = 1 \quad (3)$$

$$\text{Discordant pair (Q):} \quad \text{sgn}(s_{(x,y)}) \cdot \text{sgn}(\ell_{(x,y)}) = -1 \quad (4)$$

where, $\text{sgn}(\cdot)$ is a sign function, whose outcome is +1 when its variable is non-negative and -1 otherwise; $s_{(x,y)}$ is a pre-defined metric-based similarity function between images x and y ; $\ell_{(x,y)}$ is the annotated information difference between images x and y , i.e., $\ell_{(x,y)} = \ell_x - \ell_y$. The above approximation process can be perceived as follows. Provided image x is more similar than image y towards the query in a ranking images list, the similarity between image x and the query image (i.e., $s_{(x)}$) should be higher than that between image y and the query image (i.e., $s_{(y)}$). Also, the labeled information of image x is larger than that of image y in the ranking images list of a descending similarity order (i.e. $\ell_{(x)} > \ell_{(y)}$). According to definitions of concordant and discordant pairs, (x, y) constitutes a concordant pair, which perfectly match Equation 3 (i.e., $s_{(x,y)} = s_{(x)} - s_{(y)} > 0$ and $\ell_{(x,y)} = \ell_x - \ell_y > 0$) and P increases by 1. For the discordant pair in Equation 4, the explanation is similar.

After substituting Equations 3 and 4 into Equation 2, the original non-differentiable KT measure can be revised as Equation 5.

$$KT = \frac{1}{N_n} \sum_{x,y \in D, x \neq y} (\text{sgn}(s_{(x,y)}) \cdot \text{sgn}(\ell_{(x,y)})) \quad (5)$$

in which, D denotes all images in the ranking list. However, Equation 5 is still not ready, because the well-known step transition characteristics of sign functions exist. Therefore, the step transition problem is further tackled via the popular hyperbolic-tangent approximation way (i.e., $\text{sgn}(\xi) \approx \tanh(\xi) = \frac{\sinh(\xi)}{\cosh(\xi)} = \frac{e^{2\xi} - 1}{e^{2\xi} + 1}$, where ξ denotes the variable of corresponding functions), and a new differentiable surrogated KT measure (i.e., continuous KT, viz. CKT) can be derived in Equation 6.

$$CKT = \frac{1}{N_n} \cdot \sum_{x,y \in D, x \neq y} \left(\frac{\exp(2(s_{(x,y)})) - 1}{\exp(2(s_{(x,y)})) + 1} \cdot \frac{\exp(2(\ell_{(x,y)})) - 1}{\exp(2(\ell_{(x,y)})) + 1} \right) \quad (6)$$

Then, the gradient which is essential in SGD can be derived. After applying the chain rule, the gradient of Equation 6 can be explicitly formulated as Equation 7.

$$\begin{aligned} \nabla_{CKT}(W) &= \frac{1}{N_n} \cdot \left(\sum_{x,y \in D, x \neq y} \frac{2 \cdot \exp(2s_{(x,y)})(s_{(x,y)})' \cdot (\exp(2s_{(x,y)})) + 1}{(\exp(2s_{(x,y)})) + 1)^2} \right. \\ &\quad \left. - \frac{2 \cdot \exp(2s_{(x,y)})(s_{(x,y)})' \cdot (\exp(2s_{(x,y)})) - 1}{(\exp(2s_{(x,y)})) + 1)^2} \cdot \frac{\exp(2(\ell_{(x,y)})) - 1}{\exp(2(\ell_{(x,y)})) + 1} \right) \\ &= \frac{1}{N_n} \cdot \left(\sum_{x,y \in D, x \neq y} \frac{4 \cdot \exp(2s_{(x,y)})(s_{(x,y)})' \cdot \exp(2(\ell_{(x,y)})) - 1}{\exp(2s_{(x,y)}) + 2 \exp(2s_{(x,y)}) + 1} \cdot \frac{\exp(2(\ell_{(x,y)})) - 1}{\exp(2(\ell_{(x,y)})) + 1} \right) \\ &= \frac{1}{N_n} \cdot \left(\sum_{x,y \in D, x \neq y} \frac{4 \cdot (s_{(x,y)})'}{\exp(2s_{(x,y)}) + \exp(2s_{(y,x)}) + 2} \cdot \frac{\exp(2(\ell_{(x,y)})) - 1}{\exp(2(\ell_{(x,y)})) + 1} \right) \quad (7) \end{aligned}$$

2.2 Full-scaled Deep Metric Learning via PGD

Subsection 2.1 illustrated detailed step to derivate the gradient of CKT in the conventional SGD algorithm, and the deep metric learning can be fulfilled within the end-to-end training procedure of the whole deep learning model via back-propagation. In this subsection, an alternative more efficient PGD algorithm is introduced. Based on Equation 6, the semi-supervised deep metric learning task can be formulated via a classic LASSO (i.e., the regularization of least absolute shrinkage and selection operator) problem in Equation 8.

$$\begin{aligned} \min_W (Y - \frac{1}{N_n} \cdot \sum_{x,y \in D, x \neq y} \left(\frac{\exp(2(s_{(x,y)})) - 1}{\exp(2(s_{(x,y)})) + 1} \cdot \frac{\exp(2(\ell_{(x,y)})) - 1}{\exp(2(\ell_{(x,y)})) + 1} \right))^2 + \lambda \|W\|_1 \\ = \min_W f(W) + \lambda \|W\|_1 - I \|_1 \quad (8) \end{aligned}$$

in which, Y denotes the ideal KT value of the ranking list; $\|\cdot\|_1$ represents the $L1$ norm; $\lambda > 0$ is the regularization coefficient controlling the importance of the regularization term; and I is the

identical matrix. The loss function in Equation 8 is simplified as $f(W)$ for the ease of derivations in the following part.

It is interesting to notice that, the above LASSO problem can be considered as a speciality in the general elastic net regularization, which takes both the $L1$ norm in LASSO and the $L2$ norm in the ridge regression into consideration. Reasons to formulate the above semi-supervised deep metric learning problem into the LASSO form can be explained as follows. First, it is convenient to obtain sparse solutions in LASSO, which is beneficial towards the efficiency of this study. Second, quick solutions are feasible to tackle the LASSO problem, and the PGD algorithm is chosen here. To be specific, the solution to Equation 8 can be described as follows. Within the current k^{th} iteration of PGD, after obtaining the result W_{k-1} at the previous $(k-1)^{th}$ iteration, it is fed into Equation 9 for an intermediate variable Z_{k-1} .

$$Z_{k-1} = W_{k-1} - \frac{1}{L} \nabla f(W_{k-1}) \quad (9)$$

where, L is a positive scaling factor and ∇ denotes the gradient. Then, the popular soft thresholding technique is applied on Equation 9 for solving the i^{th} component of W at the k^{th} iteration (i.e., W_k^i) as described in Equation 10.

$$W_k^i = \text{sign}(Z_{k-1}^i - \frac{\lambda}{L}) \cdot \max\{|Z_{k-1}^i| - \frac{\lambda}{L}, 0\} \quad (10)$$

where, sign is also a sigmoid function; $\max$ represents a maximum function; Z_{k-1}^i denotes the i^{th} component of Z_{k-1} obtained via Equation 9. Furthermore, Equation 10 can be expanded via a piece-wise function in Equation 11.

$$W_k^i = \begin{cases} Z_{k-1}^i - \frac{\lambda}{L}, & \frac{\lambda}{L} < Z_{k-1}^i \\ 0, & |Z_{k-1}^i| \leq \frac{\lambda}{L} \\ Z_{k-1}^i + \frac{\lambda}{L}, & Z_{k-1}^i < -\frac{\lambda}{L} \end{cases} \quad (11)$$

In this way, the i^{th} component of W at the k^{th} iteration (i.e., W_k^i) can be quickly solved.

To sum up, the deep semi-supervised metric learning task in this study can be tackled using either conventional SGD or more efficient PGD. Both of them can be conveniently incorporated into the end-to-end learning of the whole full-scaled deep learning model, so that unknown parameters W of the metric as well as the model's parameters can be automatically determined, simultaneously.

3 EXPERIMENTAL ANALYSES

3.1 Databases and Experimental Settings

In order to evaluate the performance of the new full-scaled deep metric learning method for solving the popular re-id problem, three popular and challenging databases are incorporated for experimental analyses in this study. They are the CUHK01 database [11], the VIPeR database [6], and the QMUL-iLIDS database [27]. Details of the three databases are described as follows. For the CUHK01 database, there are totally 971 identities, each of whom has 4 different images taken by 2 non-overlapping cameras. The spatial resolution of each individual image has been consistently cropped as 160×60 . For the VIPeR database, totally 632 identities are included, and each identity in this database has 2 images taken by 2 non-overlapping cameras. All images in this database has the consistent spatial resolution of 128×48 . For the QMUL-iLIDS database, it is a database

collected from an airport during busy hours by a multi-camera CCTV system. There are 119 identities in this database, and every identity has roughly 4 images captured by 2 non-overlapping cameras. The spatial resolution of images in this database varies, and it is interesting to mention that, the database has scenarios with heavy occlusion and pose variance. The reason to incorporate the above three databases are described as follows. Databases of CUHK01 and VIPeR can be handled as a balanced learning problem, in which the exact number of images exists for each individual identity. The main difference between the VIPeR and the CUHK01 databases is that, the VIPeR database has only 2 images for each individual identity, which only accounts for half of the image number per identity in the CUHK01 database. Therefore, the balanced learning problem in the VIPeR database is more challenging, and it is also a small sample learning task, simultaneously. For the QMUL-iLIDS database, it can be viewed as an unbalanced learning problem. Hence, the re-id capability of the new full-scaled deep metric learning method will be comprehensively analyzed with respect to both the balanced learning and the unbalanced learning situations in this study.

Except for the new full-scaled deep metric learning method (i.e., annotated as "FullScale"), there are 9 other popular metric learning methods implemented for comparisons in this study. They include 3 deep learning-based methods (i.e., DRML+CKT, Norm+X-Corr [16], and Fused [1]) and 6 shallow learning-based methods (i.e., BPNN, GLCM+SVM, GLCM+DCA [9], GLCM+RCA [2], GLCM+NCA [4], and GLCM+LMNN [18]). The motivation to compare FullScale with DRML+CKT is that, DRML+CKT executes the same steps as FullScale, only except that DRML+CKT has a narrow structure. Therefore, the effect of width and cardinality in FullScale can be investigated via the comparison between FullScale and DRML+CKT. For Norm+X-Corr and Fused, they are two recently proposed deep learning-based methods for solving the popular re-id problem [16], [1]. Both of them are reported to be effective in dealing with serious problems (i.e., illumination variations, partial occlusions, wide view point changes, etc.) in the re-id task. The main difference between Norm+X-Corr and Fused is that, Norm+X-Corr applies inexact matching techniques over a wide searching space for solving the re-id problem, while Fused utilizes both inexact matching techniques as well as exact matching techniques to reduce occasional false matches caused by the wide searching space inherited in Norm+X-Corr [16], [1]. For BPNN, it is a conventional 3-layer neural network using back-propagation. For GLCM+SVM / DCA / RCA / NCA / LMNN, all of them are shallow learning-based methods and they incorporate the well-known grey-level co-occurrence matrix (i.e., GLCM) as the hand-crafted low-level feature. The reason to implement shallow learning-based methods in this experiment is to highlight the superiority of deep learning-based methods in solving the re-id problem in this study.

3.2 Results and Statistical Analyses

In Figures 5, 6, and 7, precision-and-recall curves of the re-id performance in databases of CUHK01, VIPeR, and QMUL-iLIDS are delineated, respectively. Generally speaking, the area under each precision-and-recall curve denotes the mean average precision (i.e., MAP) of the re-id performance of each individual method. It can be observed that, in Figures 5 and 6, the precision-recall curves of

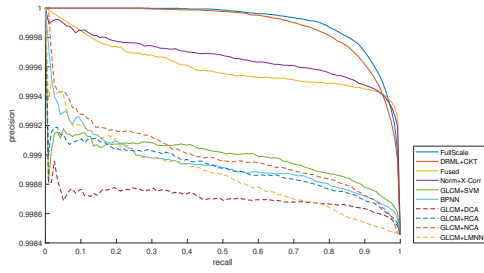


Figure 5: Precision-recall curves of all compared methods based on the CUHK01 database.

FullScale are always higher than those of others, which indicates that the new method is superior to handle the balanced learning problem in this re-id task. In Figure 7, the precision-and-recall curve of FullScale is not the highest, which reveals that other methods may become more capable to deal with the unbalanced learning problem in this re-id task.

Another more detailed quantitative analysis is conducted from the statistical perspective based on F-measures calculated from precision and recall outcomes from Figures 5, 6, and 7. Two ways of statistical evaluations, including the single-value estimation and the 95% confidence interval estimation, are included. For the single-value estimation, it is calculated as the direct difference between FullScale and each individual compared method. It turns out that, all single-value estimation results in databases of CUHK01 and VIPeR are positive, it is clear that FullScale outperforms all other compared methods in these two datasets. For the 95% confidence interval estimation, it generates an interval that the previous single-value estimation result is likely to be included (i.e., no less than 95% confidence from the statistical perspective). Statistical analyses turn out that, most upper and low bounds of 95% confidence intervals in databases of CUHK01 and VIPeR are still positive. Therefore, the superiority of FullScale when tackling the re-id task in case of the balanced learning problem in databases of CUHK01 and VIPeR can be substantiated through the above comprehensive analysis from the statistical point of view. For the problem of the unbalanced learning in the QMUL-iLIDS database, Figure 7 reveals that Norm+X-Corr and DRML+CKT can perform better than FullScale. The above qualitative observation can be further substantiated from corresponding statistical outcomes that the single-value estimation is negative and the 95% confidence interval mainly reside in the negative part. Therefore, applying inexact matchings over a wide searching space (i.e., Norm+X-Corr) or a narrow yet deep structure (i.e., DRML+CKT) may help to deal with the unbalanced learning problem in the re-id task of this study better.

4 CONCLUSION

When affective computing is emphasized on pedestrian targets within a network of cameras with non-overlapping fields of view, the challenging multi-camera-based affective computing task appears, and the pedestrian target re-identification problem becomes pre-requisite. In this study, a novel deep discriminant model-based metric learning method has been proposed to tackle the popular

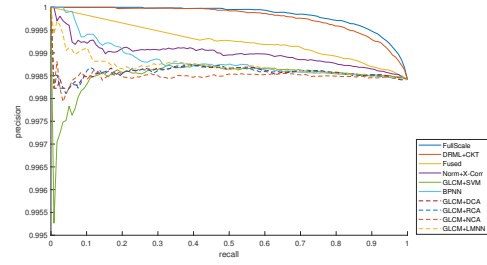


Figure 6: Precision-recall curves of all compared methods based on the VIPeR database.

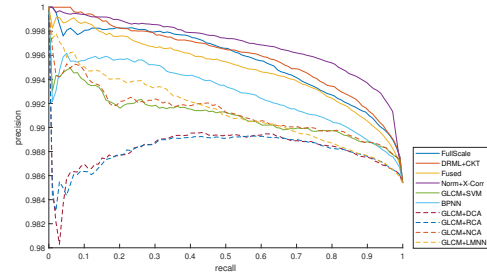


Figure 7: Precision-recall curves of all compared methods based on the QMUL-iLIDS database.

re-id problem, and there are two main novelties. First, a fully-scaled deep learning model, which simultaneously takes concepts of depth, width, and cardinality into consideration, is newly proposed. Second, the re-id task is taken from the ranking perspective, and a new rank-based deep metric learning method based on the previous fully-scaled deep learning model is introduced. This fully-scaled deep metric learning method can be realized within the end-to-end training of the fully-scaled deep learning model, and a conventional SGD algorithm as well as an alternative more efficient PGD algorithm are both introduced for its realizations. For experimental analyses, the full-scaled deep metric learning method has been compared with 9 other re-id methods from both deep learning and shallow learning perspectives based on 3 well-known and challenging databases, i.e., CUHK01, VIPeR, and QMUL-iLIDS. Comprehensive statistical analyses turn out that, the new method outperforms other compared deep learning-based and shallow learning-based methods when dealing with the balanced learning problem in the re-id task. Future efforts will be emphasized on boosting the performance of the new deep metric learning method when dealing with the challenging unbalanced learning problem in the re-id task.

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